

Experiment 03

STUDENT UTILITY PROGRAM

Title: Student Utility Program using Switch Case, Conditional Statements and Loops

Objective / Aim

To design and implement a menu-driven Student Utility Program in Java using Scanner, switch-case statements, conditional statements, nested switch statements, and looping constructs. The program performs student information, mathematical, grading, day, multiplication, summation, Fibonacci, factorial, prime number, maximum, and circle-area operations.

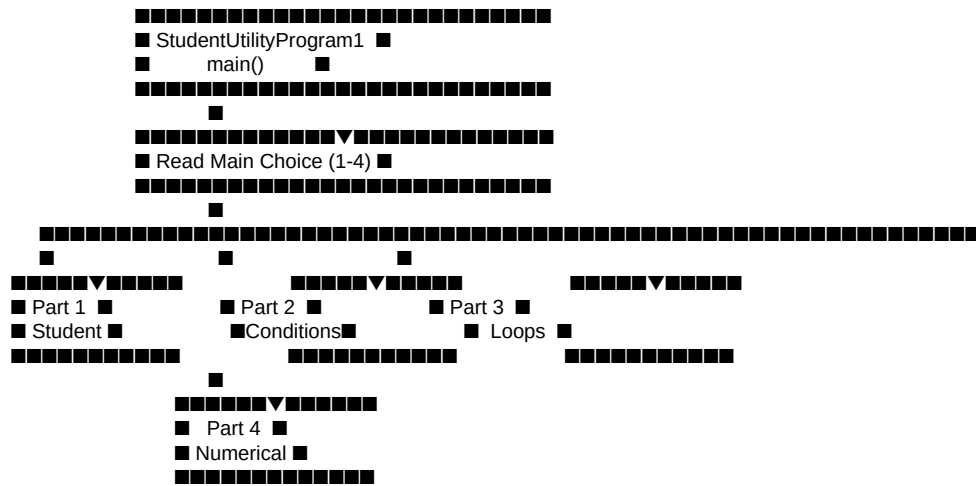
System Requirements

Hardware	Software
Standard PC/Laptop	Windows / Linux / macOS
Intel Core i3/i5 or equivalent	JDK 17 or JDK 21
Minimum 4 GB RAM	Java compiler (javac) and runtime (java)
Minimum 500 MB free storage	Eclipse / IntelliJ IDEA / NetBeans / VS Code

Algorithm / Logical Description

1. Import Scanner and create a Scanner object for keyboard input.
2. Display four main parts and read the user's choice.
3. Use an outer switch to select Part 1, Part 2, Part 3, or Part 4.
4. Use a nested switch in Part 1 for student information, details, total/percentage, and result.
5. Use Part 2 for even/odd, largest of three, grade, and day selection.
6. Use Part 3 for multiplication table, numbers 1 to N, sum, and Fibonacci series.
7. Use Part 4 for factorial, prime checking, maximum of two numbers, and circle area.
8. Use if-else statements for comparisons and conditions.
9. Use for loops for repeated calculations and series generation.
10. Display the result and invalid-choice messages where applicable.
11. Close the Scanner after execution.

Program Structure / Flow



Part 1: Student Information / Details / Total & Percentage / Result

Part 2: Even-Odd / Largest Three / Grade / Day

Part 3: Multiplication Table / 1 to N / Sum / Fibonacci

Part 4: Factorial / Prime / Maximum Two / Circle Area

Source Code

```
import java.util.Scanner;

public class StudentUtilityProgram1 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("1. Part 1");
        System.out.println("2. Part 2");
        System.out.println("3. Part 3");
        System.out.println("4. Part 4");

        System.out.print("Enter your choice: ");
        int choice = sc.nextInt();

        switch (choice) {
            case 1:
                System.out.print("Enter case 1-4: ");
                int achoice = sc.nextInt();
                String name;
                int rollno;
                double marks1, marks2, marks3, total, percentage;

                switch (achoice) {
                    case 1:
                        System.out.println("Student Information");
                        System.out.println("Name : Himanusha");
                        System.out.println("Roll No : 101");
                        break;
                    case 2:
                        sc.nextLine();
                        System.out.print("Enter Student Name: ");
                        name = sc.nextLine();
                        System.out.print("Enter Roll Number: ");
                        rollno = sc.nextInt();
                        System.out.print("Enter English Marks: ");
                        marks1 = sc.nextDouble();
                        System.out.print("Enter Maths Marks: ");
                        marks2 = sc.nextDouble();
                        System.out.print("Enter Social Marks: ");
                        marks3 = sc.nextDouble();
                        System.out.println("\n----- STUDENT DETAILS -----");
                        System.out.println("Name : " + name);
                        System.out.println("Roll No : " + rollno);
                        System.out.println("English Marks : " + marks1);
                        System.out.println("Maths Marks : " + marks2);
                        System.out.println("Social Marks : " + marks3);
                        break;
                    case 3:
                        System.out.print("Enter English Marks: ");
                        marks1 = sc.nextDouble();
                        System.out.print("Enter Maths Marks: ");
                        marks2 = sc.nextDouble();
                        System.out.print("Enter Social Marks: ");
                        marks3 = sc.nextDouble();
                        total = marks1 + marks2 + marks3;
                        percentage = (total / 300) * 100;
                        System.out.println("Total = " + total);
                        System.out.println("Percentage = " + percentage);
                        break;
                    case 4:
                        sc.nextLine();
                        System.out.print("Enter Student Name: ");
                        name = sc.nextLine();
                        System.out.print("Enter Roll Number: ");
                        rollno = sc.nextInt();
                        System.out.print("Enter English Marks: ");
                        marks1 = sc.nextDouble();
                        System.out.print("Enter Maths Marks: ");
                        marks2 = sc.nextDouble();
                        System.out.print("Enter Social Marks: ");
                        marks3 = sc.nextDouble();
                        total = marks1 + marks2 + marks3;
                        percentage = (total / 300) * 100;
                        System.out.println("\n----- RESULT -----");
                        System.out.println("Name : " + name);
                        System.out.println("Roll No : " + rollno);
                        System.out.println("Total : " + total);
```

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        System.out.println("Percentage : " + percentage);
        break;
    default:
        System.out.println("Invalid Choice");
    }
    break;

case 2:
    System.out.print("Enter case 1-4: ");
    int bchoice = sc.nextInt();
    switch (bchoice) {
        case 1:
            System.out.print("Enter a number: ");
            int n = sc.nextInt();
            if (n % 2 == 0) System.out.println("Even Number");
            else System.out.println("Odd Number");
            break;
        case 2:
            System.out.print("Enter first number: ");
            int a = sc.nextInt();
            System.out.print("Enter second number: ");
            int b = sc.nextInt();
            System.out.print("Enter third number: ");
            int c = sc.nextInt();
            if (a >= b && a >= c) System.out.println("Largest = " + a);
            else if (b >= a && b >= c) System.out.println("Largest = " + b);
            else System.out.println("Largest = " + c);
            break;
        case 3:
            System.out.print("Enter percentage: ");
            double per = sc.nextDouble();
            if (per >= 90) System.out.println("Grade A");
            else if (per >= 75) System.out.println("Grade B");
            else if (per >= 60) System.out.println("Grade C");
            else if (per >= 40) System.out.println("Grade D");
            else System.out.println("Fail");
            break;
        case 4:
            System.out.print("Enter day number (1-7): ");
            int day = sc.nextInt();
            switch (day) {
                case 1: System.out.println("Monday"); break;
                case 2: System.out.println("Tuesday"); break;
                case 3: System.out.println("Wednesday"); break;
                case 4: System.out.println("Thursday"); break;
                case 5: System.out.println("Friday"); break;
                case 6: System.out.println("Saturday"); break;
                case 7: System.out.println("Sunday"); break;
                default: System.out.println("Invalid Day");
            }
            break;
        default:
            System.out.println("Invalid Choice");
    }
    break;

case 3:
    System.out.print("Enter case 1-4: ");
    int cchoice = sc.nextInt();
    switch (cchoice) {
        case 1:
            System.out.print("Enter a number: ");
            int num = sc.nextInt();
            System.out.println("Multiplication Table:");
            for (int i = 1; i <= 10; i++)
                System.out.println(num + " x " + i + " = " + (num * i));
            break;
        case 2:
            System.out.print("Enter N: ");
            int n1 = sc.nextInt();
            System.out.println("Numbers from 1 to " + n1 + ":");
            for (int i = 1; i <= n1; i++) System.out.print(i + " ");
            System.out.println();
            break;
        case 3:
            System.out.print("Enter N: ");
            int m = sc.nextInt();
            int sum = 0;
            for (int i = 1; i <= m; i++) sum = sum + i;
            System.out.println("Sum = " + sum);
            break;
        case 4:

```

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        System.out.print("Enter number of terms: ");
        int terms = sc.nextInt();
        int x = 0, y = 1, z;
        System.out.println("Fibonacci Series:");
        for (int i = 1; i <= terms; i++) {
            System.out.print(x + " ");
            z = x + y;
            x = y;
            y = z;
        }
        System.out.println();
        break;
    default:
        System.out.println("Invalid Choice");
    }
    break;

case 4:
    System.out.print("Enter case 1-4: ");
    int dchoice = sc.nextInt();
    switch (dchoice) {
        case 1:
            System.out.print("Enter a number: ");
            int factNum = sc.nextInt();
            long fact = 1;
            for (int i = 1; i <= factNum; i++) fact = fact * i;
            System.out.println("Factorial = " + fact);
            break;
        case 2:
            System.out.print("Enter a number: ");
            int primeNum = sc.nextInt();
            if (primeNum < 2) {
                System.out.println("Not a Prime Number");
                break;
            }
            int count = 0;
            for (int i = 1; i <= primeNum; i++)
                if (primeNum % i == 0) count++;
            if (count == 2) System.out.println("Prime Number");
            else System.out.println("Not a Prime Number");
            break;
        case 3:
            System.out.print("Enter first number: ");
            int first = sc.nextInt();
            System.out.print("Enter second number: ");
            int second = sc.nextInt();
            if (first > second) System.out.println("Maximum = " + first);
            else System.out.println("Maximum = " + second);
            break;
        case 4:
            System.out.print("Enter radius: ");
            double r = sc.nextDouble();
            double area = 3.14 * r * r;
            System.out.println("Area of Circle = " + area);
            break;
        default:
            System.out.println("Invalid Choice");
    }
    break;

default:
    System.out.println("Invalid Choice");
}
sc.close();
}
}

```

Output

Example 1:

1. Part 1

2. Part 2

3. Part 3

4. Part 4

Enter your choice: 2

Enter case 1-4: 3

Enter percentage: 82

Grade B

Example 2:

Enter your choice: 3

Enter case 1-4: 4

Enter number of terms: 7

Fibonacci Series:

0 1 1 2 3 5 8

Completion and Execution Commands

1. Save the program as **StudentUtilityProgram1.java**.
2. Open Command Prompt / Terminal and move to the folder containing the file.
3. Compile: **javac StudentUtilityProgram1.java**
4. Execute: **java StudentUtilityProgram1**
5. Enter the main-part choice and then the corresponding case choice.

Result / Completion

The Student Utility Program was successfully designed in Java. It demonstrates Scanner input, switch-case and nested switch-case statements, if-else conditions, for loops, arithmetic operations, series generation, and menu-driven program execution.